Introduction to Data Analytics

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Disclaimer

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Topics Covered

- What is Data Analytics?
- Why is Data Analytics Important?
- The Data Analytics Process
- Types of Data Analytics
- Tools and Technologies in Analytics
- Scope and Career
- Application of Data Analytics

Topics Covered

- ❑ What is Business Intelligence?
- ❑ Why Businesses use BI?
- ❑ Business Intelligence vs Data analytics
- ❑ Key Components of BI
- ❑ Architecture and Workflow

Topics Covered

- ❑ Overview of Power BI Desktop
- ❑ Key Features of Power BI Desktop
- ❑ PowerBI workflow
- ❑ Installing and Setting up Power BI Desktop

What is Data Analytics?

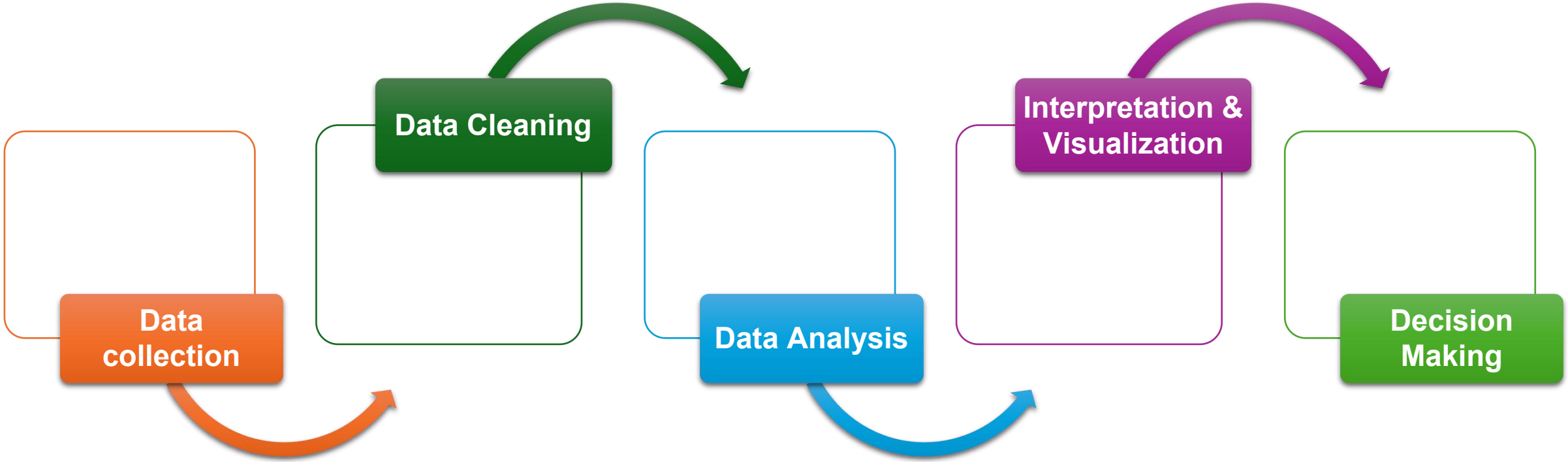
- Data Analytics is the **systematic analysis of data** to extract meaningful insights.
- It involves **collecting, cleaning, processing, and interpreting data** to support decision-making.
- It helps businesses **understand trends, patterns**, and behaviors.

Source: https://hevodata.com/learn/ultimate-guide-to-data-analytics/

Why is Data Analytics Important?

- Drives informed decision - making
- Identifies Business Opportunities and in efficiencies
- Strategic planning and predictive insights
- Customer Experience
- Competitive advantage

The Data Analytics Process



Types of Data Analytics

Type	Question Answered	Purpose	Techniques Used	Examples
Descriptive Analytics	What happened?	Summarize past data to understand trends and patterns	Data aggregation, data mining, reports, dashboards	Monthly sales report, website traffic summary
Diagnostic Analytics	Why did it happen?	Investigate the root causes of outcomes	Drill-down, data discovery, correlations	Sales dropped due to supply chain issues
Predictive Analytics	What could happen?	Forecast future trends and outcomes	Statistical modeling, machine learning, regression	Predicting customer churn or stock prices
Prescriptive Analytics	What should we do?	Recommend actions based on predictions	Optimization, simulation, decision trees	Dynamic pricing strategy, route optimization

Tools and Technologies in Analytics

Power BI

Tableau

Excel

Python & R

SQL

Scope of Data Analytics

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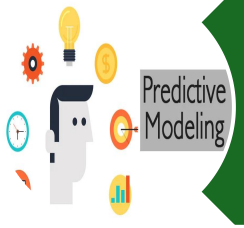
What is the Scope of Data Analytics?

- The scope of data analytics refers to the **range of applications, industries, roles, skills, and future opportunities** associated with collecting, analyzing, and interpreting data to support decision-making, improve performance, and drive innovation across organizations.

Scope of Data Analytics

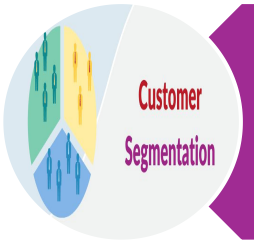
Applications of Data Analytics

Business reporting & dashboards



Predictive modeling for sales, risk, or demand

Process optimization in operations



Customer segmentation & Personalization

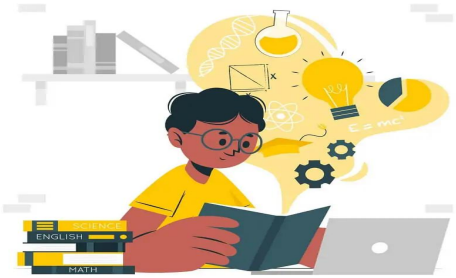
Industries Leveraging Data Analytics

Health care

Finance

Retail

Logistics



Education

Sports

Career Roles in Data Analytics

- Data Analyst
- BI Developer
- Data Scientist
- Data Engineer

Skills Required in Data Analytics

Technical Skills

- SQL
- **Power BI**
- Excel

Soft Skills

- Communication
- Business Acumen
- Problem solving

Overlap

- Insight driven thinking
- Story telling with data

Future Opportunities in Data Analytics

AI-Powered insights

Cloud based analytics

Self – service BI tools for non tech users

Real time & edge analytics

Ethical data use privacy Compliance

Business Intelligence Concepts

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What is Business Intelligence?

- **Business intelligence (BI**) is a set of technological processes **for collecting, managing and analyzing organizational data** to yield insights that inform business strategies and operations.

Source: https://images.app.goo.gl/6N5y5QfWFjKkHE4cA

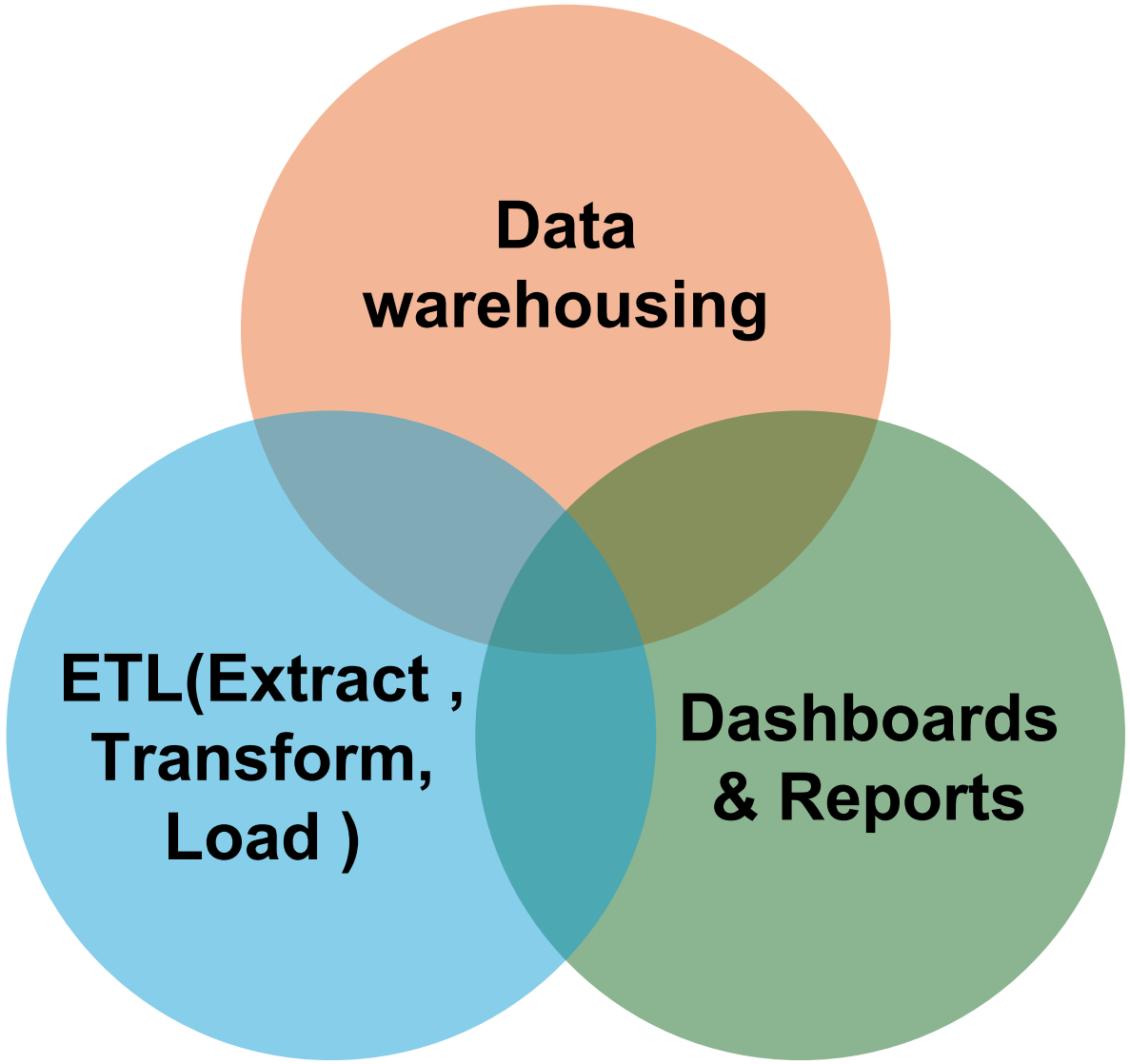
Why Businesses Use BI

- Understand performance trends
- Faster data driven decisions
- Identify opportunities , risks, and bottlenecks
- Monitor KPI and optimize operations

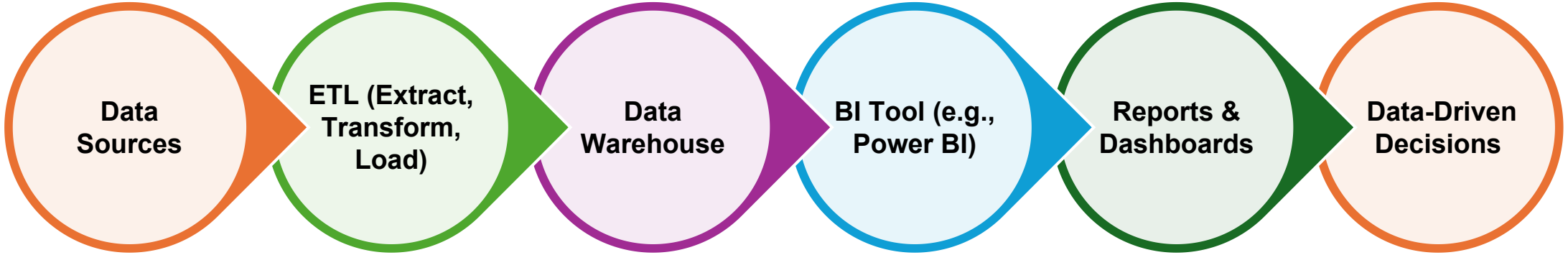
Business Intelligence vs. Data Analytics

Feature	Business Intelligence(BI)	Data Analytics
Focus	What happened?	Why it happened, what could happen?
Tools	Power BI, Tableau	Python, R, SQL
Output	Dashboards, KPIs	Predictive models, deep analysis
Users	Business Users, Executives	Data Analysts, Data Scientists

Key Components of Business Intelligence



BI Architecture & Workflow



Common BI Tools

Power BI **Tableau** **Qlik**

Overview of Power BI Desktop

What is Power BI?

- Microsoft's Business Intelligence & data visualization tool
- Converts data into interactive reports & dashboards
- No/low-code, user-friendly for both tech & non-tech users

Power BI

Why Choose Power BI?

- User – friendly Interface
- Strong Microsoft Integration
- Interactive Dashboards
- Cost effective
- AI & Advanced features
- Self service & Enterprise ready

Power BI Product Family

Product	Purpose
Power BI Desktop	Design & build reports on your Windows PC
Power BI Service	Share, publish, and collaborate on reports online
Power BI Mobile	Access dashboards and reports on mobile devices

Power BI Desktop

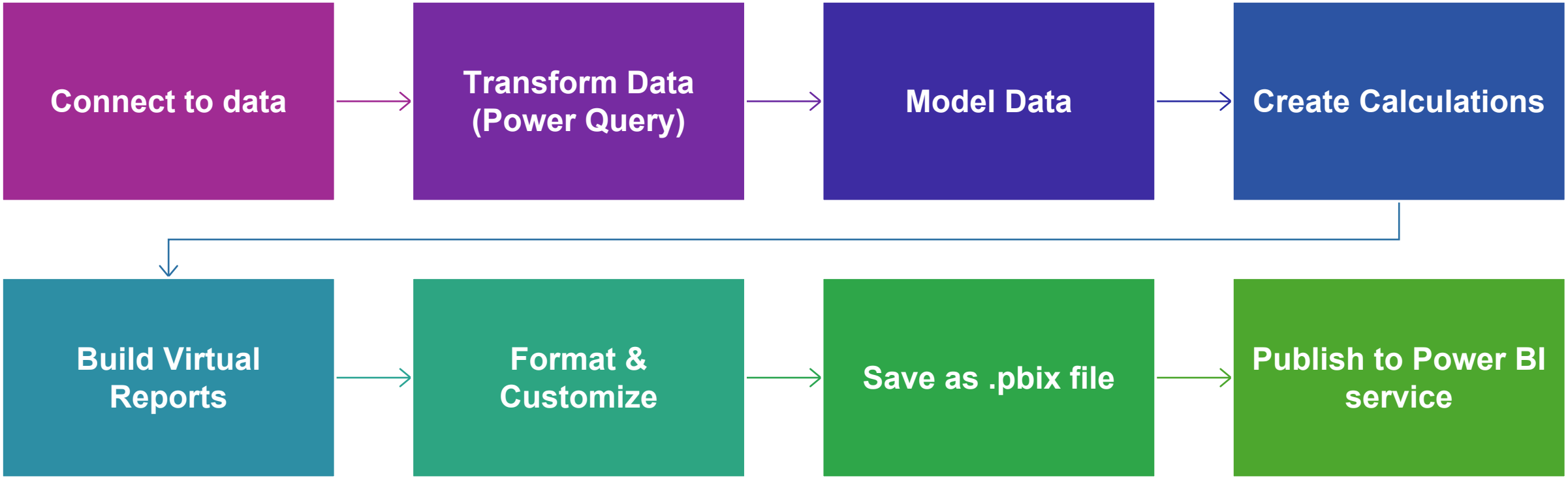
- Free Windows application to **create data reports**
- Connects to data sources **(Excel, SQL, web, cloud)**
- Used to **clean, transform, model, and visualize data**
- Saves files as **.pbix** for sharing or publishing
- Core tool in the **Power BI ecosystem** for report building

Source: https://images.app.goo.gl/wABsEupKPtsDHoYD9

Key Features of Power BI Desktop

- Connecting to multiple data services
- Clean & transform data
- Create data models
- Design interactive reports & dashboards
- Use DAX for calculations

Power BI Desktop Workflow



Installing & setting up Power BI

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Download Microsoft Power BI Desktop from Official ...

Install Instructions. **Download** the version of **Power BI Desktop** that matches the architecture (x86 or x64) of your Windows OS. Run the EXE installer and follow ...

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Microsoft Power BI Desktop

Microsoft Power BI Desktop is built for the analyst. It combines state-of-the-art interactive visualizations, with industry-leading data query and modeling built-in. Create and publish your reports to Power BI. Power BI Desktop helps you empower others with timely critical insights, anytime, anywhere.

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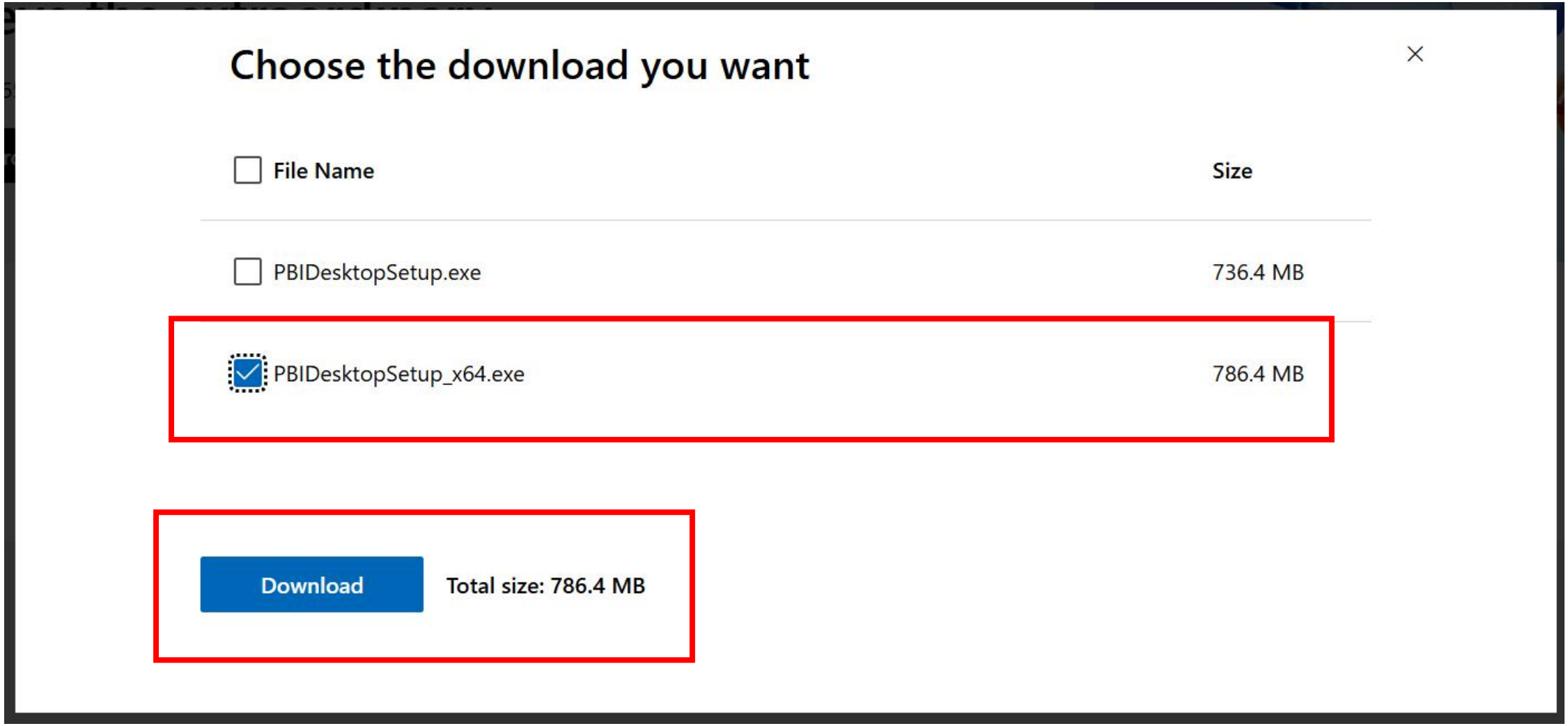

Select PBIDesktopSetup_x64.exe , click on Download

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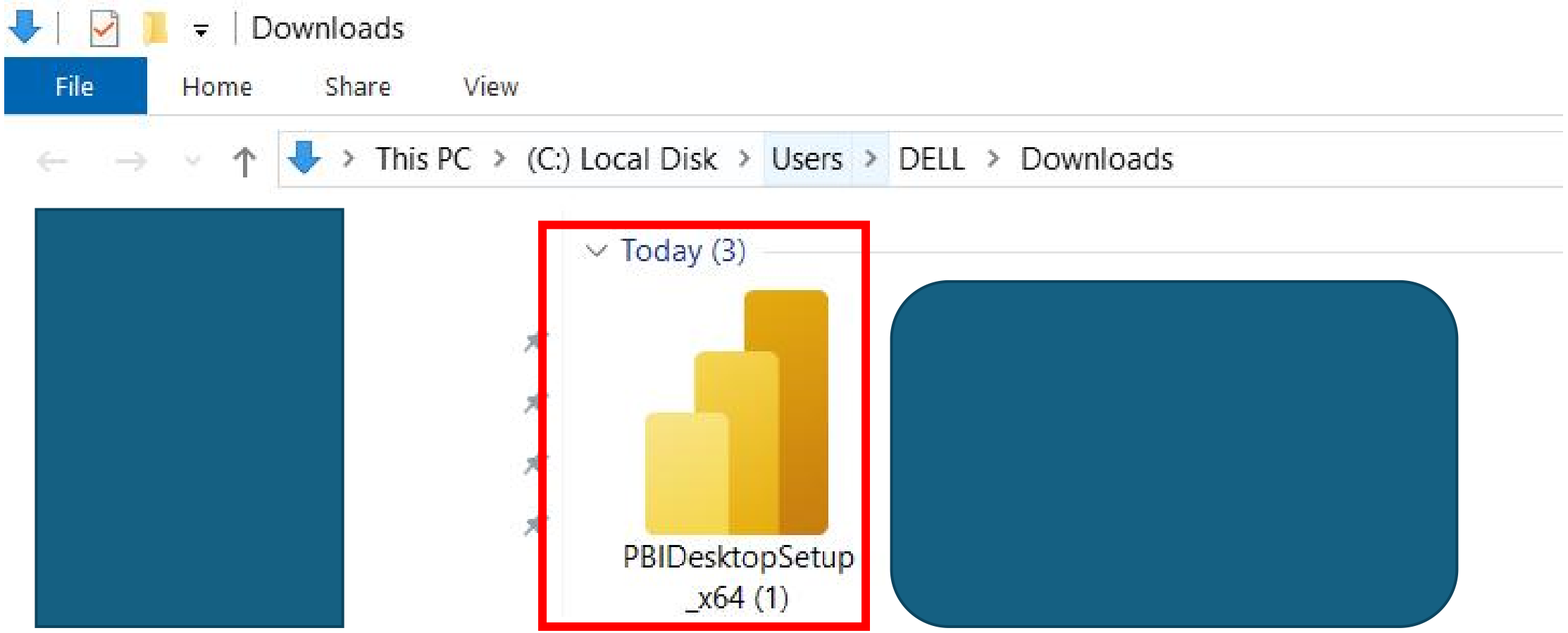
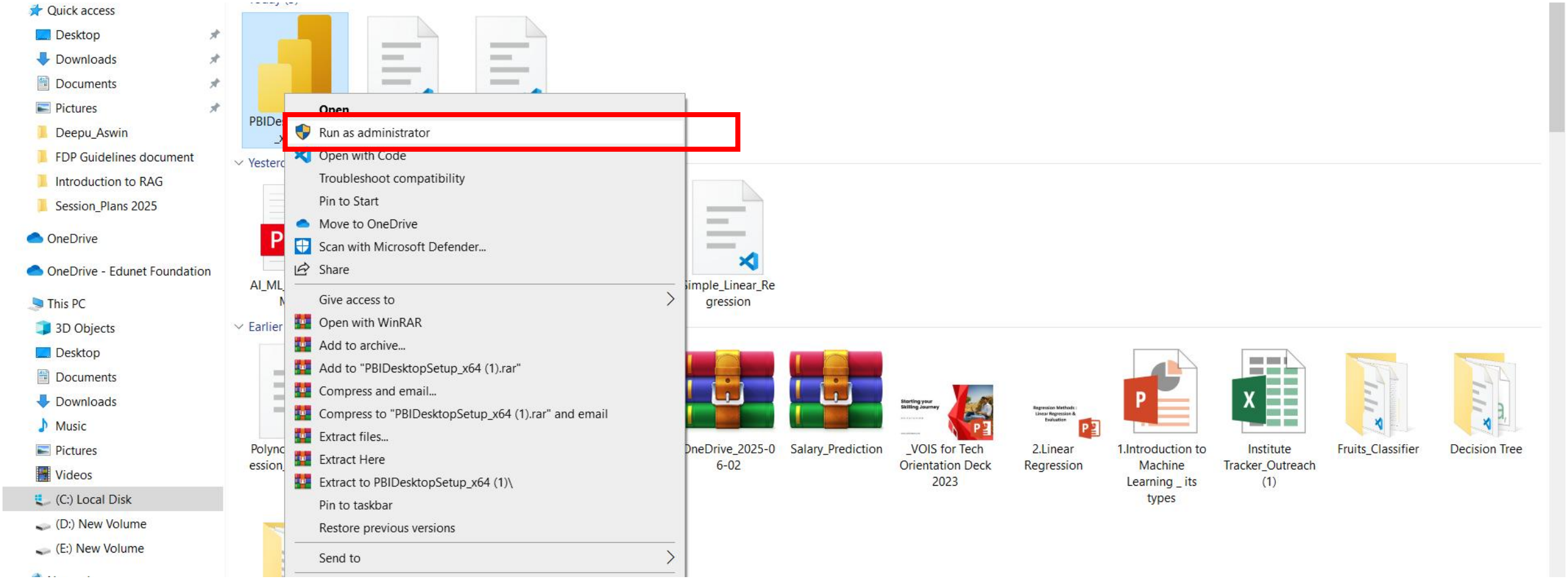
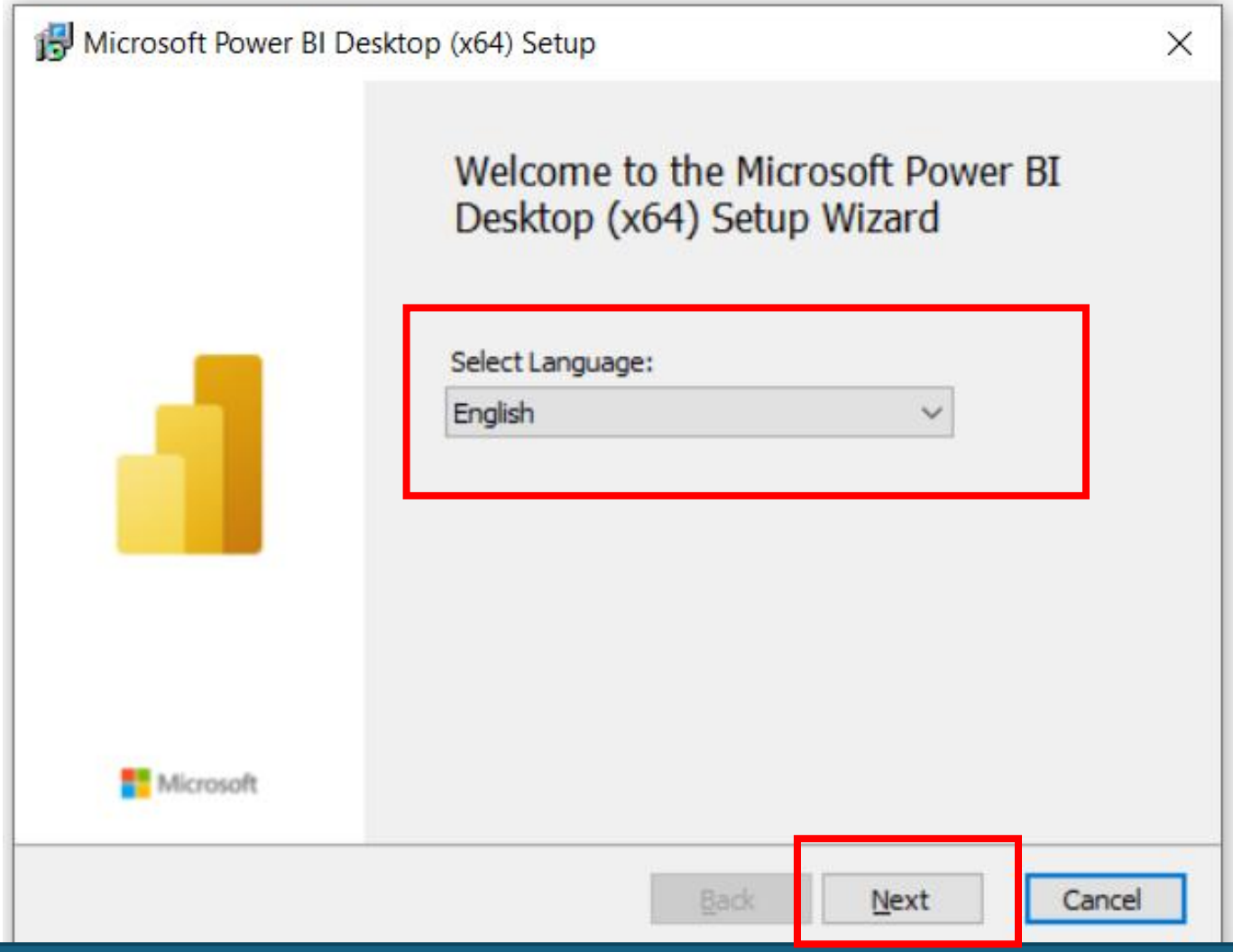


Go to the download folder, click on PBIDesktopSetup_x64



Right click on the icon , select Run as administrator



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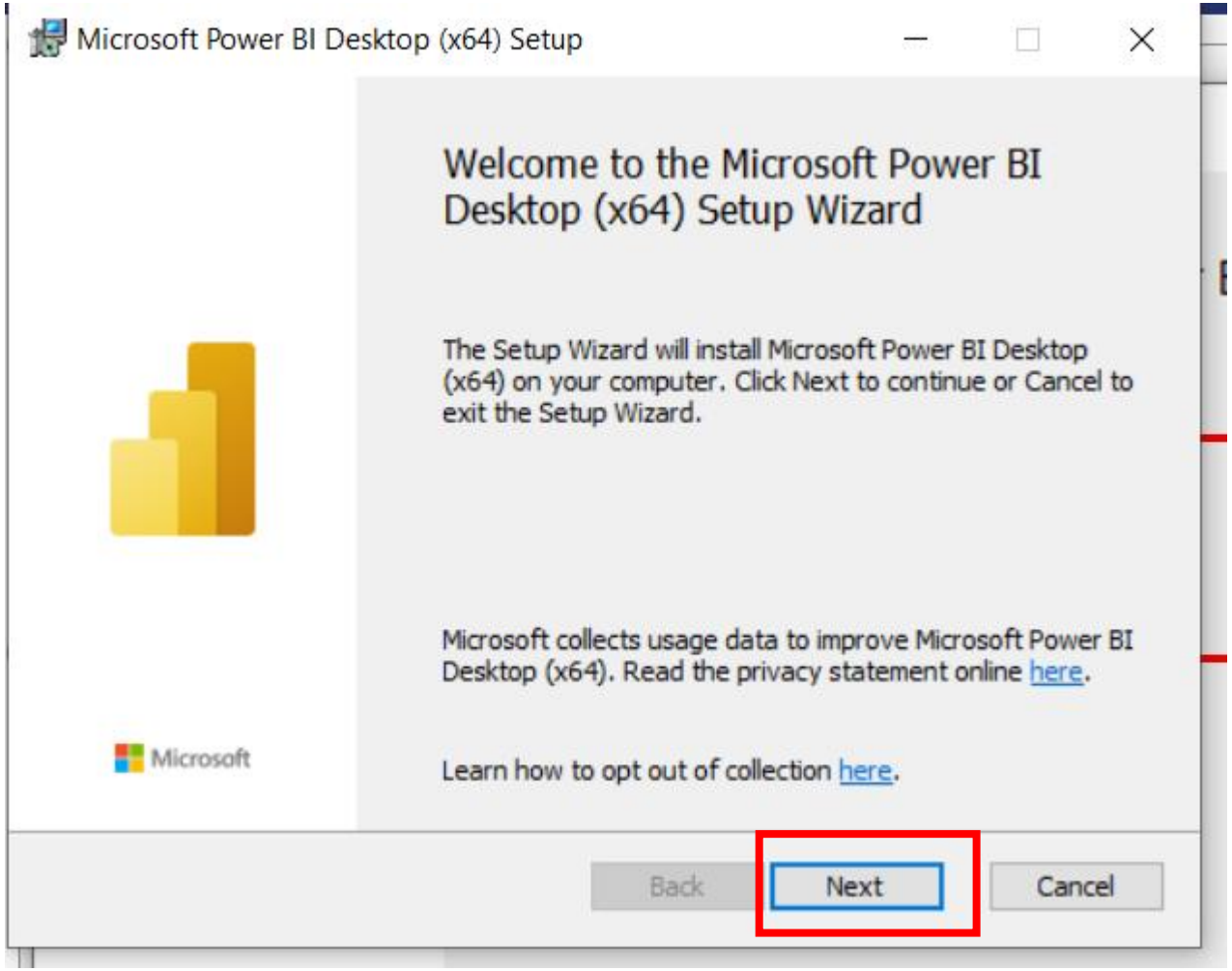

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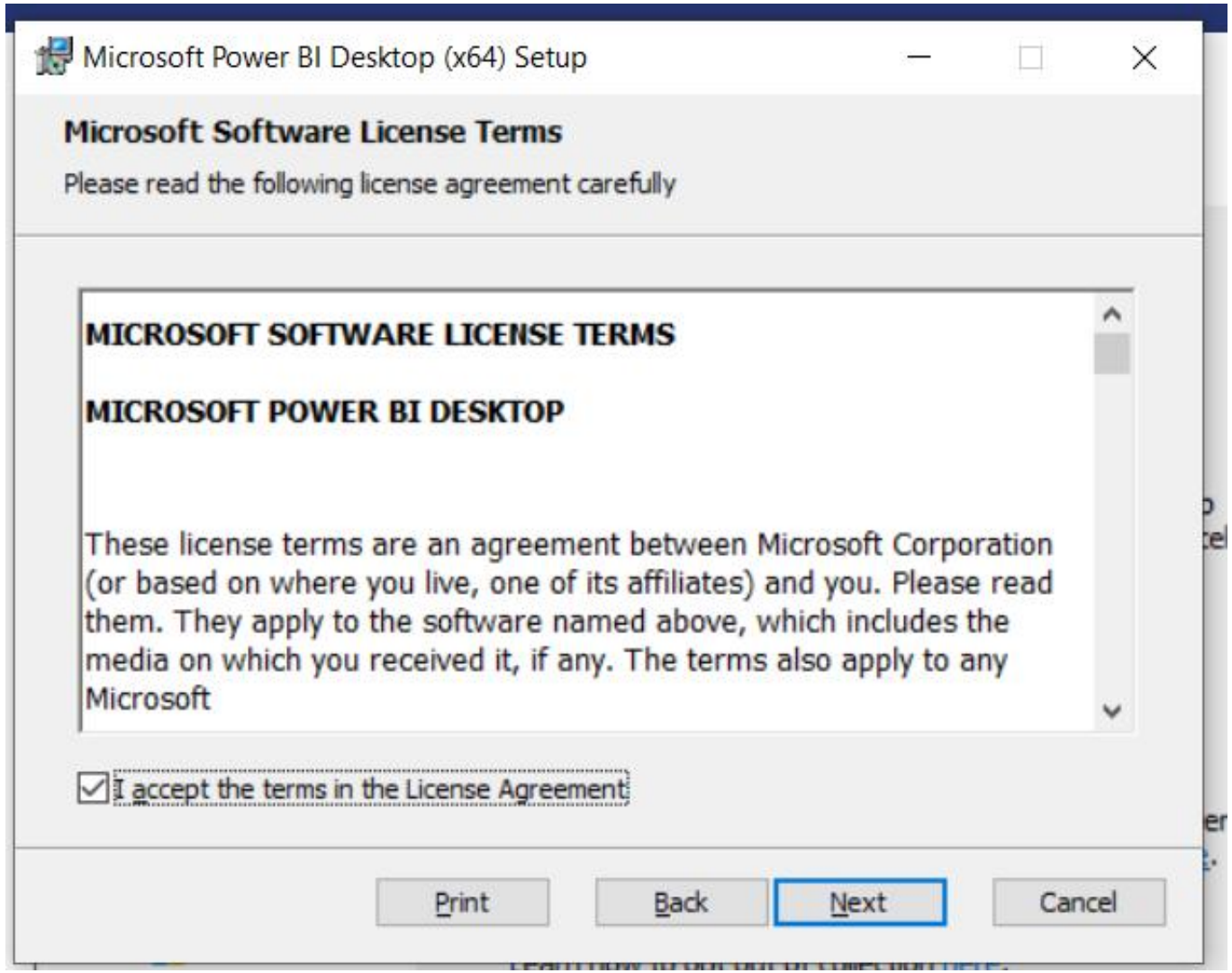

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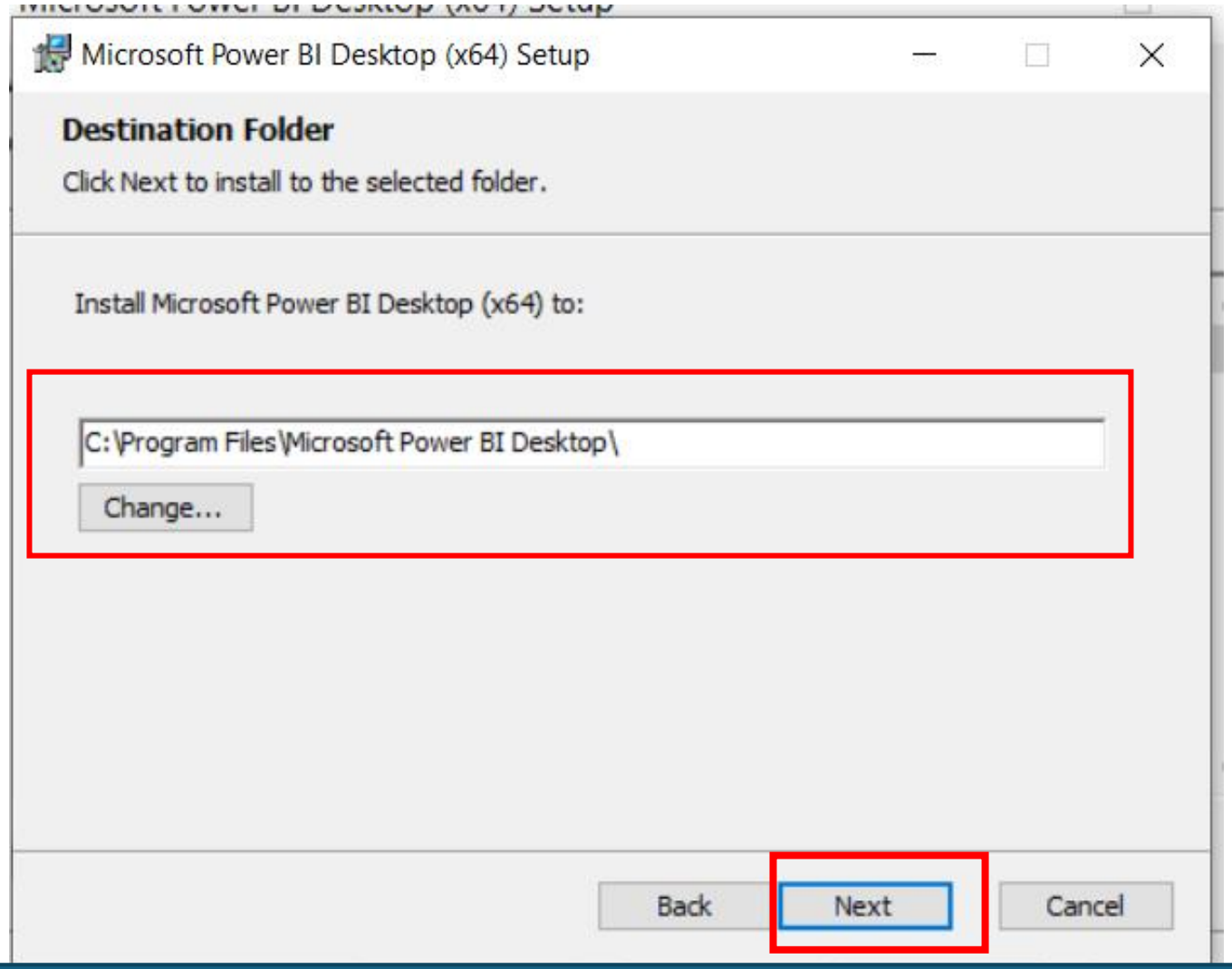

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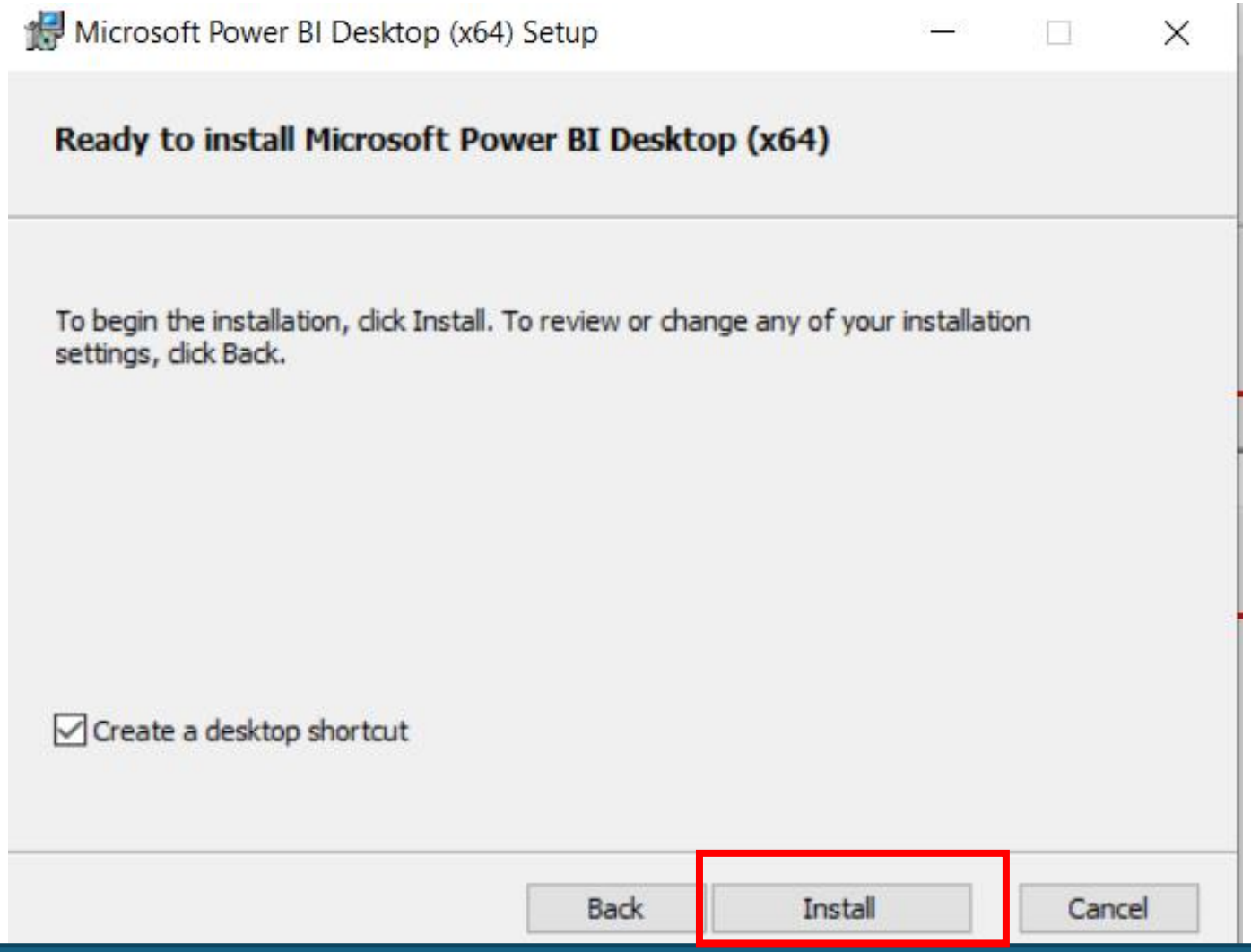
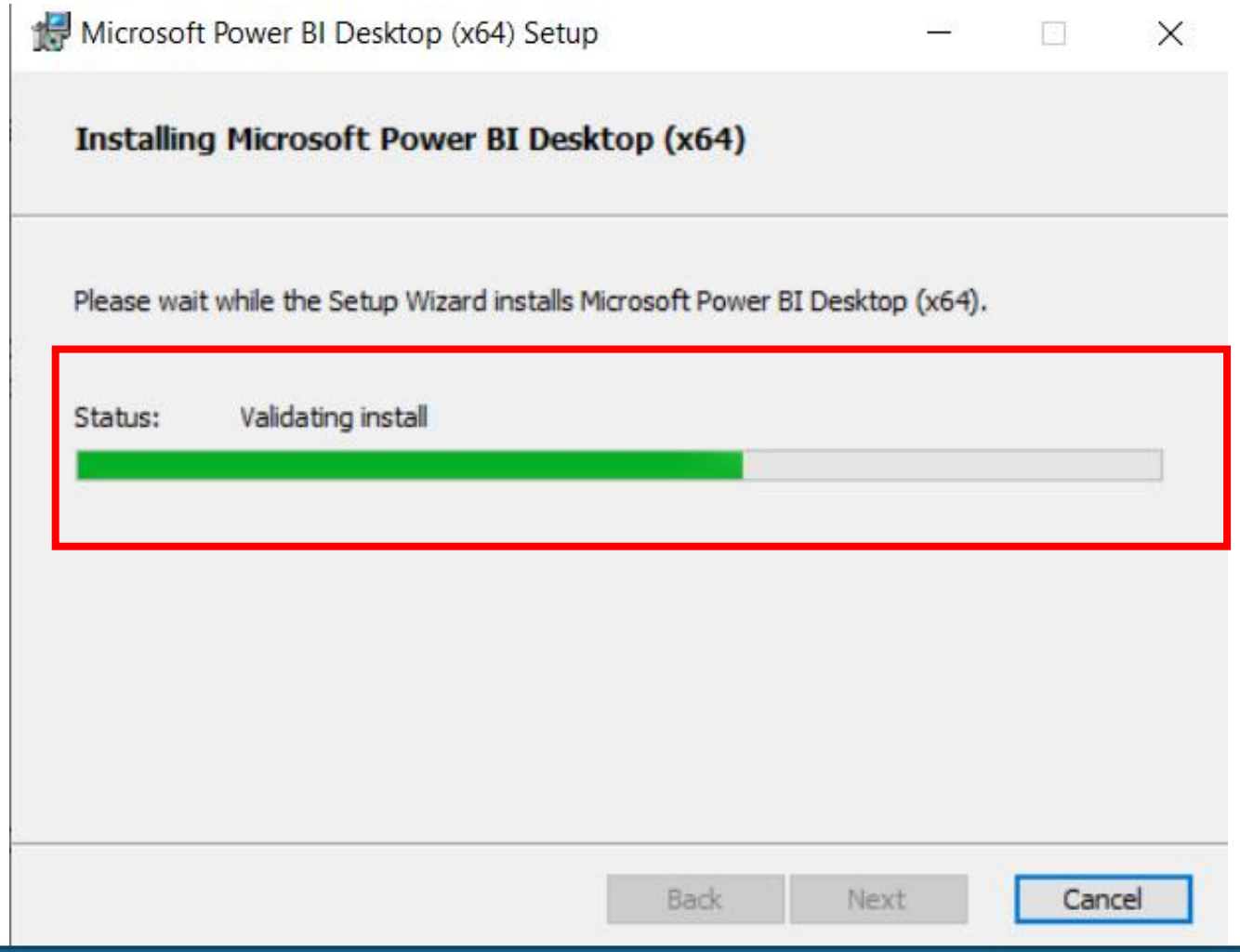
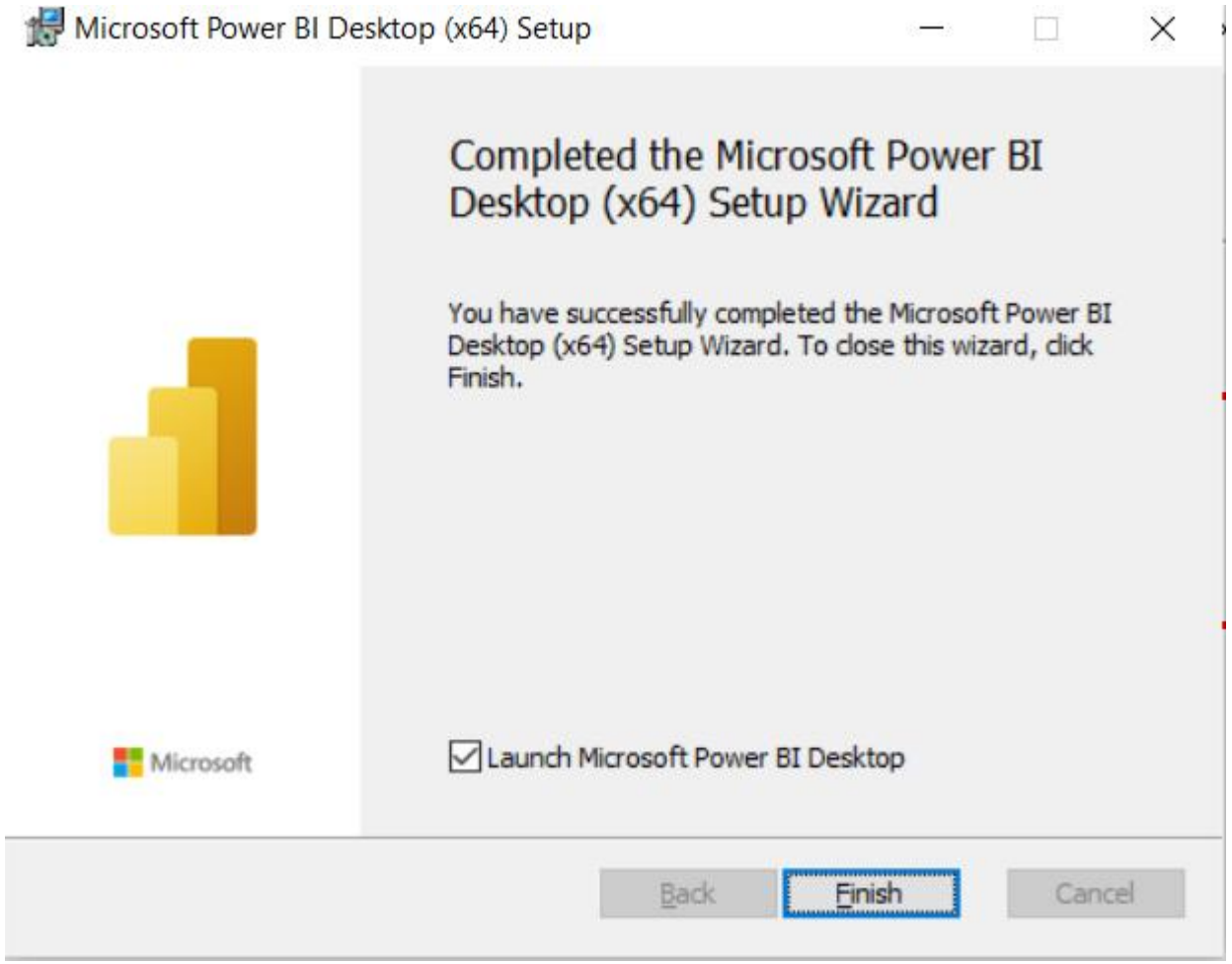


Click on Install


Microsoft Power BI Desktop (x64) Setup
Installing Microsoft Power BI Desktop (x64)
Please wait while the Setup Wizard installs Microsoft Power BI Desktop (x64).
Status:
Validating install
Back
Next
Cancel

Installing….



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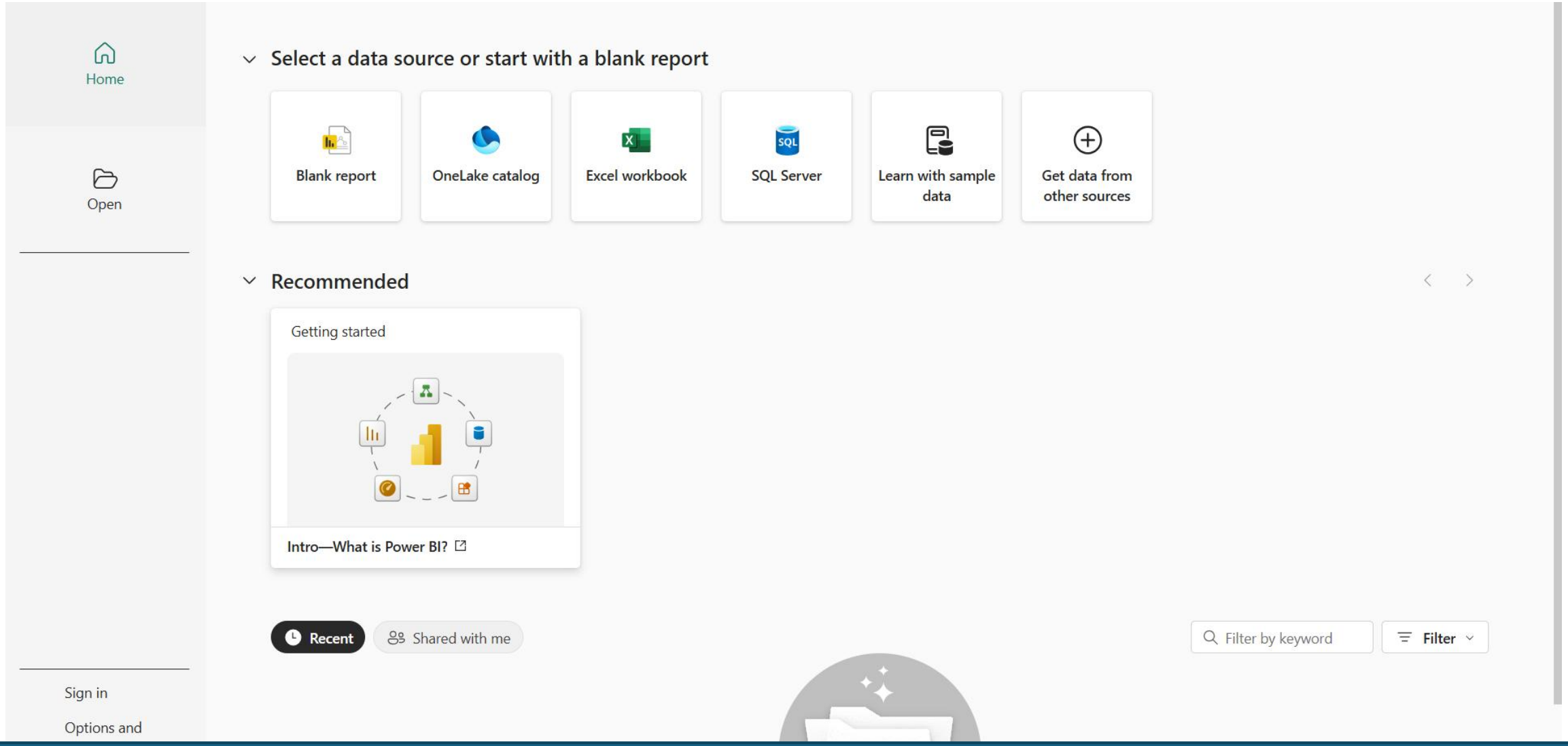
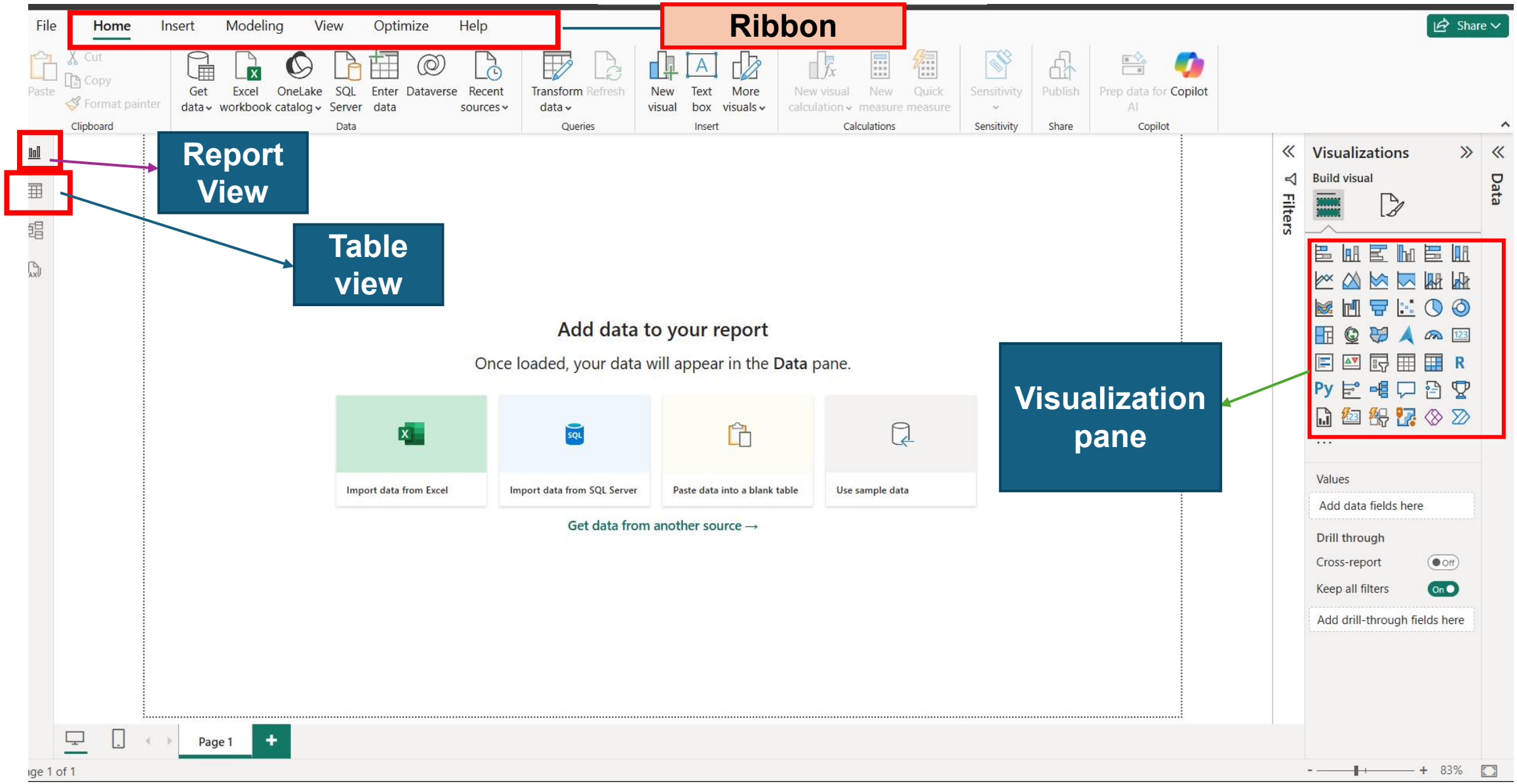


Power BI installed successfully!!

Power BI Desktop Interface Overview



QnA

Thank you!